BeatAware-Radar2ECGNet: Physics-Constrained Continuous ECG Reconstruction from Radar

Xiao Jin[1(✉)], Sira Yongchareon[1], and Anuradha Singh[1]

[1] Department of Data Science and AI, School of Engineering, Computer and Mathematical Sciences, Auckland University of Technology, Auckland, New Zealand
xiao.jin@autuni.ac.nz, sira.yongchareon@aut.ac.nz, anuradha.singh@aut.ac.nz

Abstract. Contactless ECG reconstruction from radar offers a promising solution for unobtrusive cardiac monitoring but remains a challenging cross-modal learning problem. Radar measures thoracic mechanical motion, while ECG records cardiac electrical activity, making the mapping indirect, subject-dependent, and temporally misaligned. We propose BeatAware-Radar2ECGNet, a physiology-aware framework for continuous radar-to-ECG waveform reconstruction under subject-independent evaluation. Instead of treating the task as a direct black-box regression, the proposed model constrains the reconstruction process through kinematic inversion, cardiac prior conditioning, temporal convolutional modeling, and physical alignment. Kinematic inversion extracts compact motion channels from multi-bin radar inputs; cardiac prior conditioning injects QRS/P/T structural cues via auxiliary supervision and TFiLM modulation; and physical alignment compensates for latent electromechanical delay before ECG decoding. Experiments on the public MMECG dataset with an 11-fold leave-one-subject-out protocol show that BeatAware-Radar2ECGNet achieves a PCC of 0.772, an RMSE of 0.096, a QMR of 99.9%, and an R-F1 score of 0.946. It also obtains lower fiducial timing errors than representative continuous-output baselines, with Q-Q, R-R, S-S, and T-T errors of 15.2 ms, 8.20 ms, 12.7 ms, and 11.5 ms, respectively. Compared with the TCN backbone, the proposed model improves PCC from 0.683 to 0.772 and reduces RMSE from 0.111 to 0.096, demonstrating that physiological priors and latent temporal alignment improve not only waveform fidelity but also beat-level monitoring reliability.

Keywords: Radar-based ECG reconstruction · Contactless cardiac monitoring · FMCW radar · Cross-modal learning · Physiological prior modeling

1 Introduction

Cardiovascular diseases remain a major global health burden, motivating the development of continuous, unobtrusive cardiac monitoring outside clinical settings. An electrocardiogram (ECG) is a central diagnostic signal because it directly records cardiac electrical activity and reveals beat timing, rhythm, and

morphology [11,31]. However, conventional ECG acquisition relies on electrodes attached to the skin, which may cause discomfort, skin irritation, and reduced compliance during long-term monitoring [24]. Contactless sensing with radio-frequency and millimeter-wave radar offers an appealing alternative: a radar placed near the user can measure subtle chest-wall motion induced by respiration and heartbeat without body contact [7,19,24].

Despite its strong clinical promise, the continuous reconstruction of clinically informative ECG-like waveforms from radar signals remains a challenging, inherently ill-posed cross-modal learning problem. Unlike HR or RR estimation, waveform-level ECG reconstruction requires preserving fine-grained P–QRS–T morphology and beat-to-beat temporal consistency [14]. The main bottleneck arises from the electromechanical domain gap: radar measures external thoracic mechanical motion, whereas ECG records internal cardiac electrical activity [4,32]. Although these two modalities are physiologically coupled through cardiac electromechanical processes, the mapping from radar signals to ECG waveforms is indirect, nonlinear, non-stationary, and not uniquely determined [16,29]. In practice, radar measurements are also vulnerable to respiration, body movements, multipath reflections, and ambient clutter, which can obscure the weak millimeter-level cardiac motion [7,15,19,21]. This ill-posed mapping is further exacerbated by strong inter-subject variability caused by anatomical differences, posture changes, and radar-to-body distance [4,14,29,32]. As a result, standard black-box regressors often fail under subject-independent settings, leading to R-peak shifts, temporal misalignment, and degraded reconstruction of low-amplitude P and T waves.

Our empirical evaluations indicate that strict leave-one-subject-out (LOSO) radar-to-ECG reconstruction faces severe generalization bottlenecks under cross-subject settings. When evaluated under strict cross-subject isolation, where the target individual's data is completely excluded from the training phase, severe individual variation and domain shift cause conventional networks to produce anatomically distorted waveforms. To overcome these cross-subject discrepancies without relying on subject-specific adaptation windows, historical calibration data, or few-shot test-time optimization, recent advancements have shifted toward incorporating explicit control-theoretic state space constraints [29,30], cross-modal prior knowledge transfer [16], and multi-scale semantic feature analysis [14] to decouple subject-specific mechanical fluctuations from generalized cardiac electrical rhythms effectively.

We propose BeatAware-Radar2ECGNet to address this cross-modal challenge under strict subject isolation. As illustrated in Fig. 1, radar-based ECG reconstruction suffers from an inherent electromechanical domain gap, since radar captures thoracic mechanical motion while ECG records cardiac electrical activity. The main contribution of this work is a physiology-aware reconstruction architecture that explicitly couples radar kinematic inversion, cardiac-prior-guided temporal conditioning, and physical alignment. Specifically, Kinematic Inversion (KI) learns compact cardiac motion channels from the 50-bin FMCW radar range-time signal; Cardiac Prior Conditioning (CP) uses auxiliary QRS/P/T

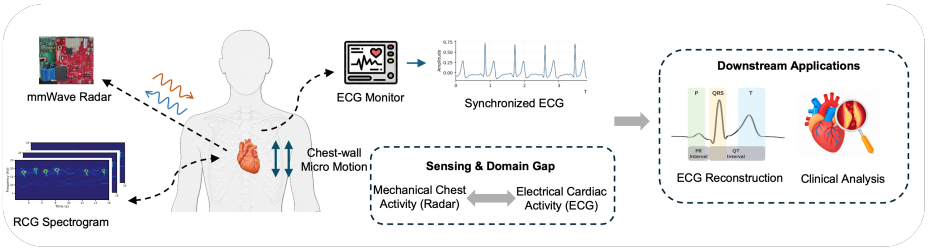


Fig. 1. Overview of radar-based contactless ECG reconstruction.

structural cues and rhythm embeddings to modulate temporal features; and Physical Alignment (PA) compensates for latent, subject-dependent electromechanical delays prior to ECG decoding. Evaluated on the public MMECG dataset with an 11-fold leave-one-subject-out protocol, the proposed framework shows more balanced waveforms, fiducial timing, and monitoring performance than representative continuous-output baselines. The main contributions of this work are summarized as follows:

- We propose a physiology-aware end-to-end framework that combines kinematic inversion, cardiac-prior conditioning, temporal convolutional modeling, and physical alignment for continuous radar-to-ECG reconstruction.
- We introduce cardiac-prior conditioning and physical alignment modules to guide beat-aware morphology reconstruction and compensate for latent electromechanical delay in the radar-to-ECG mapping.
- We conduct an 11-fold LOSO evaluation on the public MMECG dataset, demonstrating improved RMSE, fiducial timing accuracy, QMR, and R-F1 over representative continuous-output baselines.

2 Related Work

2.1 Contactless Cardiac Monitoring with Radar

Early radar-based vital sign tracking primarily targeted macro-level metrics, such as respiration rate (RR), heart rate (HR), or basic heartbeat intervals, using continuous-wave (CW) or ultra-wideband (UWB) signals processed via classical filters and frequency-domain analysis [9,22]. Recently, high-resolution frequency-modulated continuous-wave (FMCW) millimeter-wave radar has shifted the focus toward granular, segment-level cardiac event characterization. For instance, CardiacWave pioneered high-definition heart activity computing by exploiting the cardiorespiratory scattering effect to recover full continuous waves [25]. Similarly, UWB systems have been configured for ECG-grained monitoring by tracking short-range spatial phase updates [22].

To enhance downstream diagnostic value, recent work has moved beyond sinus rhythms to investigate complex cardiac pathologies and fiducial features. Cao et al. demonstrated the non-contact extraction of key ECG fiducial points

from radar signals using multi-neighbor differentiation to define semantic boundaries [2]. Furthermore, distribution-encoding networks paired with microwave biomedical radars have been successfully validated for multi-class arrhythmia screening, achieving robust statistical consensus across diverse patient cohorts [3]. Although these studies show that radar signals contain useful cardiac-related information, they also highlight a major limitation: raw radar observations are easily disturbed by respiration and body movements, leading to noticeable waveform distortion.

2.2 Deep Cross-Modal ECG Reconstruction

Driven by the limitations of linear filtering, deep learning has become the primary paradigm for bridging the mechanical-to-electrical gap. Rather than relying on explicit biomechanical modeling, recent frameworks utilize high-capacity neural networks to learn cross-modal mapping functions directly. Early attempts processed short-time Fourier transform (STFT) spectrograms through residual spectral-spatial encoder-decoder frameworks (RSSRNet) to decouple respiratory and cardiac components from chest-wall motion [10, 23]. Similarly, simple convolutional recurrent autoencoders (CRAE) built with bidirectional LSTM layers have been configured to model long-term temporal dependencies across raw radar sequences [13].

To constrain this inherently ill-posed radar-to-ECG mapping, recent investigations have introduced explicit architectural or physical priors. One prominent direction embeds mathematical constraints directly into the network. For example, radarODE and radarODE-MTL incorporate ordinary differential equations into the decoder backbone to govern the morphological evolution of cardiac states, utilizing multitask learning and eccentric gradient alignment to handle sensor noise and movement artifacts [29, 30]. Another mainstream category leverages generative frameworks to handle complex cross-domain discrepancies. AirECG introduces a cross-domain conditional diffusion architecture to synthesize multi-lead waveforms [32]. Similarly, ECGDiff and other advanced temporal diffusion layers treat radar cardiography (RCG) as a structural condition, iteratively denoising noisy sequences into clinical-grade electrical trajectories to recover sharp R-peaks and interval boundaries [6, 28]. Other generative alternatives utilize multi-scale bottleneck fusion, cross-modal semantic distillation from heart sounds, or attention-driven generative adversarial networks (Attention-DCGAN) to reconstruct stable ECG sequences from clustered heartbeat patterns [8, 12, 14]. To mitigate the domain dissonance further, Liu et al. introduced a cascade cross-modal autoencoder that transfers representations from a pre-trained ECG prior to a radar processor, minimizing deep feature losses to restrict morphological deviations [17]. Despite their strong modeling capacity, existing regression-based methods often treat radar-to-ECG reconstruction as a direct mapping problem. However, this mapping is highly sensitive to posture changes, anatomical variability, and subject-specific sensing conditions, which can lead to unstable waveform reconstruction.

3 Method

Problem Formulation: We formulate radar-to-ECG reconstruction as a cross-modal sequence learning task. Let $\mathcal{D} = \{(X_i, Y_i, s_i)\}_{i=1}^{N}$ denote the paired dataset, where X_i is an FMCW radar segment, Y_i is the synchronized ECG waveform, and s_i is the subject identity. Each radar input is a range-time window $X_i \in \mathbb{R}^{C_{\text{in}} \times T}$, with $C_{\text{in}} = 50$ range bins and $T = 1600$ samples. At 200 Hz, each window covers 8 s, and the ECG target is $Y_i \in \mathbb{R}^{1 \times T}$.

Given a mini-batch $X \in \mathbb{R}^{B \times C_{\text{in}} \times T}$, the goal is to learn a mapping

$$\hat{Y} = f_\theta(X), \qquad \hat{Y} \in \mathbb{R}^{B \times 1 \times T}, \tag{1}$$

where $\hat{Y}$ is the reconstructed ECG and θ denotes trainable parameters. We evaluate the model under a leave-one-subject-out (LOSO) protocol, where training, validation, and test subjects are strictly disjoint.

Since radar measures thoracic mechanical motion while ECG records cardiac electrical activity, the mapping is indirect and ill-posed. We therefore decompose f_θ as

$$M = \mathcal{F}_{\text{KI}}(X), \quad P = \mathcal{F}_{\text{CP}}(M), \quad H = \mathcal{F}_{\text{TCN}}(M \mid P), \quad \hat{Y} = \mathcal{G}_{\text{dec}}(\mathcal{F}_{\text{PA}}(H)). \tag{2}$$

Here, KI extracts compact radar motion features, CP provides cardiac priors, the TCN backbone models temporal dynamics, PA compensates latent electromechanical delay, and the decoder reconstructs the ECG waveform. The overall architecture is shown in Fig. 2.

Overall BeatAware-Radar2ECGNet Architecture: As shown in Fig. 2, the proposed network consists of Kinematic Inversion (KI), Cardiac Prior Conditioning (CP), a TCN-based temporal backbone, Physical Alignment (PA), and an ECG decoder. Given the radar input $X \in \mathbb{R}^{B \times 50 \times 1600}$, KI first converts the multi-bin range-time representation into compact motion channels, and CP further predicts soft cardiac prior maps:

$$M = \text{Conv}_{1\times1}(\alpha \odot X), \qquad P = \mathcal{F}_{\text{CP}}(M), \tag{3}$$

where α denotes the learned range-bin attention weights, $\odot$ is element-wise multiplication, and $M \in \mathbb{R}^{B \times 16 \times 1600}$. This step suppresses redundant range-bin responses while retaining radar-derived motion cues related to cardiac activity. The design is motivated by the fact that cardiac-related micro-motion is usually concentrated in a limited subset of informative range bins, whereas other bins may contain respiration-dominant responses, multipath reflections, or background clutter. Therefore, the attention-projection operation serves as a data-driven kinematic compression mechanism that reduces redundant spatial responses before temporal ECG reconstruction.

The predicted prior map P provides coarse temporal cues for ECG structures, such as QRS-, P-, and T-related regions. During training, P is supervised by prior masks derived from the ECG, whereas during inference, it is predicted solely from radar features. Unlike unconstrained regression, the auxiliary prior

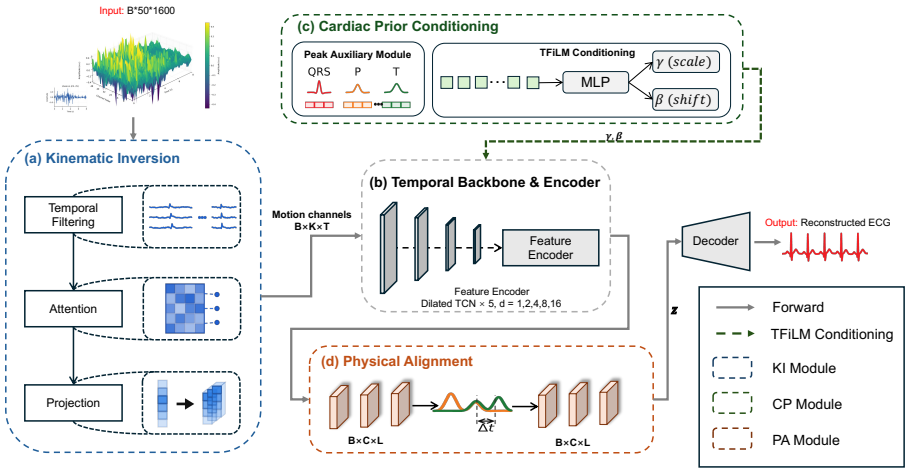


Fig. 2. Architecture of the proposed BeatAware-Radar2ECGNet. The model reconstructs continuous ECG waveforms from FMCW radar range-time inputs through four main components: (a) Kinematic Inversion (KI), which converts 50 range bins into compact motion channels; (b) a TCN-based temporal feature extractor for latent cardiac dynamics modeling; (c) Cardiac Prior Conditioning (CP), which predicts QRS/P/T priors and generates TFiLM modulation parameters; and (d) Physical Alignment (PA), which compensates for latent electromechanical delay before ECG decoding. Solid arrows denote the main reconstruction flow, while dashed arrows indicate prior-guided TFiLM conditioning.

branch encourages the latent representation to preserve beat-structured temporal landmarks. This is particularly important for radar-to-ECG reconstruction because low-amplitude ECG components, such as P and T waves, may be weakly observable in radar measurements but remain temporally coupled with the dominant QRS rhythm.

To inject the cardiac prior into temporal feature learning, CP transforms P into feature-wise modulation parameters and applies them to the backbone features:

$$\tilde{H} = H \odot (1 + \gamma(P)) + \beta(P), \tag{4}$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are learnable projections. This modulation follows the principle of feature-wise affine conditioning [20], allowing beat-level prior information to adaptively modulate temporal representations. As a result, the network is encouraged to align reconstructed ECG morphology with the underlying cardiac rhythm rather than relying only on direct point-wise regression.

The conditioned representation is then processed by a lightweight TCN backbone. A temporal stem first projects the motion features to $B \times 128 \times 400$ using a 1D convolution with a kernel size of 7 and a stride of 4. Five residual TCN blocks with dilation factors $\{1, 2, 4, 8, 16\}$ are then used to model multi-scale temporal dependencies:

$$H_l = H_{l-1} + \mathcal{T}_l(H_{l-1}), \tag{5}$$

where $\mathcal{T}_l(\cdot)$ denotes two dilated convolution layers with normalization, activation, and dropout. Dilated temporal convolutions provide a large receptive field with efficient parallel computation, making them suitable for long-range sequence modeling [1]. In this work, the increasing dilation enlarges the temporal receptive field without recurrent computation, enabling efficient modeling of local ECG morphology and beat-to-beat dynamics.

Before decoding, PA compensates the latent delay between radar-observed mechanical motion and ECG electrical activity using a learnable channel-wise temporal filter:

$$Z_c(t) = \sum_{\tau=-d}^{d} a_c(\tau) H_c(t-\tau), \tag{6}$$

where $a_c(\tau)$ is the alignment kernel for channel c and d controls the temporal support. The PA module is designed to account for the latent electromechanical delay between cardiac electrical activation and mechanically observable thoracic motion. Instead of imposing a fixed global delay, the proposed channel-wise temporal filter allows the model to estimate local temporal compensation in the latent feature space. Finally, the decoder upsamples Z from length 400 to 1600 through two transposed convolution blocks and a final 1D projection layer, producing the reconstructed ECG waveform $\hat{Y} \in \mathbb{R}^{B\times 1\times 1600}$.

Training Objective: The model is trained with a joint objective that combines waveform fidelity, spectral consistency [27], and cardiac-prior supervision:

$$\mathcal{L} = \mathcal{L}_1(\hat{Y}, Y) + \lambda_{\text{stft}}\mathcal{L}_{\text{STFT}}(\hat{Y}, Y) + \lambda_{\text{cp}}\text{BCE}(P, \bar{P}), \tag{7}$$

where $\mathcal{L}_1$ measures point-wise waveform reconstruction error, $\mathcal{L}_{\text{STFT}}$ denotes the multi-resolution spectral loss, and P and $\bar{P}$ are the predicted and ECG-derived QRS/P/T cardiac-prior maps. The PA module is optimized implicitly through this objective without requiring explicit delay annotations. This objective preserves ECG morphology in both time and frequency domains while encouraging rhythm-aware temporal alignment.

4 Experiments

Experimental Setup. *Dataset and protocol.* We evaluate our model on the public MMECG dataset[1] [4], which contains synchronized FMCW radar and reference ECG recordings for radar-based ECG reconstruction. The dataset was collected using a TI AWR1843 radar operating at 77 GHz with 3.8 GHz bandwidth. It includes 91 trials from 11 healthy subjects under four physiological conditions: normal breathing (NB), irregular breathing (IB), post-exercise (PE), and sleep (SP). Both radar and ECG signals are sampled at 200 Hz. For each trial, the radar input is represented as a range-time matrix with 50 range bins. We segment the synchronized radar and ECG recordings into 8-s windows, where each input is $X \in \mathbb{R}^{50\times 1600}$ and the target ECG is $Y \in \mathbb{R}^{1\times 1600}$. We adopt an 11-fold dataset split protocol to evaluate subject-independent generalization.

[1] `https://github.com/jinbochen0823/RCG2ECG`

Baselines. We compare the proposed method with representative continuous-output baselines, including GRU, BiLSTM, Transformer, U-Net1D, TCN, WaveGRU [26], AirECG [32], RSSRNet [23], and radarODE [30]. The first five models serve as general sequence-to-sequence neural baselines, covering recurrent, attention-based, convolutional, and temporal-convolutional designs. WaveGRU, AirECG, RSSRNet, and radarODE represent recent radar-based ECG reconstruction methods with different modeling strategies, including multi-scale recurrent modeling, diffusion-based generation, spectral-spatial reconstruction, and ODE-constrained temporal modeling. All reproduced baselines are evaluated using the same window length, subject split, and metric pipeline.

Evaluation metrics. We evaluate the results using three types of metrics. Morphological metrics include PCC and RMSE, measuring waveform similarity and amplitude error. Since global waveform similarity does not necessarily guarantee accurate beat localization, we further include event-based and monitoring-oriented metrics. Event-based timing metrics include Q-Q, R-R, S-S, and T-T errors, measuring absolute timing differences between corresponding ECG fiducial points. Clinical metrics include QMR (Qualified Monitoring Rate) and R-F1@150 ms for monitoring reliability, as well as RR, PR, QRS, and QT interval errors for interval-level clinical consistency [5]. All timing errors are reported in milliseconds, interval errors are reported as relative differences, and QMR is reported as a percentage. R-peaks and fiducial points are detected using NeuroKit2 [18] for both reconstructed and reference ECG signals.

Implementation details. All models are implemented in PyTorch and trained on a single NVIDIA GeForce RTX 4080 SUPER GPU. Unless otherwise stated, models are trained for up to 150 epochs with a batch size of 16, an initial learning rate of 1×10^{-4}, and a weight decay of 1×10^{-4} using AdamW. A cosine learning-rate scheduler and an early-stopping patience of 20 epochs are used. The best checkpoint is selected according to the validation PCC. For each LOSO fold, one subject is held out for testing, one subject from the remaining subjects is used for validation, and all other subjects are used for training. Radar and ECG segments are normalized at the segment level before model training.

5 Results and Discussion

Overall Comparison. Table 1 reports the quantitative comparison on MMECG under the LOSO protocol. The proposed model achieves the best overall performance among continuous-output models, with the lowest RMSE and fiducial timing errors and the highest QMR and R-F1. Compared with the TCN backbone, it improves PCC from 0.683 to 0.772 and reduces RMSE from 0.111 to 0.096, showing the benefit of the proposed physics-aware modules.

The proposed model also provides more comprehensive performance than recent radar-based ECG reconstruction methods. Although radarODE reports a higher PCC, its fiducial and clinical outputs are unavailable in our unified evaluation pipeline. For radar-to-ECG reconstruction, PCC mainly reflects global waveform correlation and may remain high even when fiducial points are temporally shifted. Therefore, beat-level metrics such as fiducial timing errors, QMR,

Table 1. Comparison with current radar-to-ECG reconstruction methods on MMECG. The best results are highlighted in bold, and the second-best results are underlined.

Methods	Params	Morph.		Event-based Timing				Clinical	
		PCC↑	RMSE↓	Q-Q↓	R-R↓	S-S↓	T-T↓	QMR↑	R-F1↑
GRU	0.38M	0.180	0.158	25.1	24.7	25.8	26.5	52.2%	0.418
BiLSTM	0.30M	0.048	0.175	24.5	25.9	24.8	29.2	55.0%	0.499
Transformer	0.98M	0.308	0.153	22.4	17.0	20.6	27.4	82.7%	0.518
U-Net1D	0.30M	0.345	0.150	20.5	16.1	20.8	23.6	95.4%	0.764
TCN	1.3M	0.683	0.111	<u>17.2</u>	11.8	<u>15.0</u>	<u>13.9</u>	97.4%	<u>0.941</u>
WaveGRU [26]	0.49M	0.608	<u>0.125</u>	22.7	<u>10.9</u>	19.3	16.3	<u>99.4%</u>	0.854
AirECG [32]	20.1M	0.147	0.206	26.7	25.6	26.4	27.1	86.8%	0.787
RSSRNet [23]	11.8M	0.233	0.179	25.1	24.9	26.0	25.3	91.9%	0.800
radarODE [30]	35.2M	**0.870**	0.131	–	–	–	–	–	–
Ours	4.8M	<u>0.772</u>	**0.096**	**15.2**	**8.20**	**12.7**	**11.5**	**99.9%**	**0.946**

Note: Q-Q, R-R, S-S, and T-T errors are reported in milliseconds. R-F1 is computed with a 150-ms tolerance window. QMR is reported in percentage.

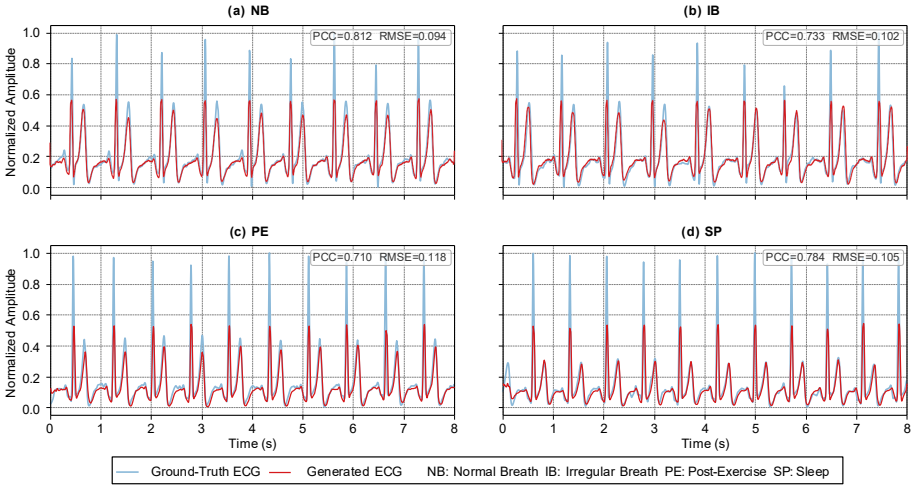


Fig. 3. Representative ECG reconstruction examples under four physiological conditions: (a) Normal Breathing (NB), (b) Irregular Breathing (IB), (c) Post-Exercise (PE), and (d) Sleep Condition (SP). Ground-truth and reconstructed ECG signals are shown after amplitude normalization; PCC and RMSE are reported for each example.

and R-F1 are necessary to assess whether the reconstructed ECG is reliable for cardiac monitoring. Under this broader evaluation, the proposed model achieves the lowest Q-Q, R-R, S-S, and T-T errors of 15.2 ms, 8.20 ms, 12.7 ms, and 11.5 ms, respectively, with a QMR of 99.9% and an R-F1 score of 0.946.

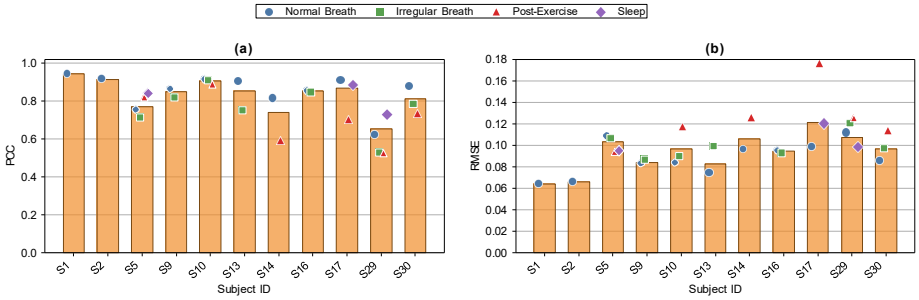


Fig. 4. Subject- and condition-wise reconstruction results. Bars denote subject-level mean values, while markers indicate condition-specific mean results.

Reconstruction and Robustness Analysis. The qualitative examples in Fig. 3 show that the proposed model reconstructs ECG-like waveforms across normal breathing, irregular breathing, post-exercise, and sleep conditions. The reconstructed signals preserve the dominant rhythm and align well with the reference R-peaks, indicating that the model captures the main cardiac motion patterns in the radar measurements. Fine P/T morphology and beat-to-beat amplitude variations remain more difficult to assess, reflecting the weak coupling between subtle ECG components and radar-observed thoracic motion. Subject- and condition-wise results in Fig. 4 further demonstrate the robustness of the proposed method. Although intersubject variability arises from differences in body geometry and radar coupling, the model maintains stable performance across most subjects. Normal breathing and sleep generally yield higher reconstruction quality, whereas post-exercise conditions remain more challenging due to a faster heart rate and stronger cardiorespiratory dynamics. These results suggest that the proposed model is robust to both subject-level and condition-level distribution shifts.

Distributional and Fiducial Timing Analysis. We further assess segment-level reliability using the CDF curves in Fig. 5. The proposed model shows a favorable distribution, with PCC shifted toward higher values and RMSE concentrated in the lower-error region. The R-R interval error curve also indicates stable beat-to-beat timing for most test segments, suggesting that the improvement is consistent rather than dominated by a few easy cases.

Fiducial timing errors are analyzed in Fig. 6. R-peak localization is the most stable among the four fiducial points, which is consistent with the stronger mechanical response around ventricular contraction. In contrast, Q and T points show larger variability because they are lower-amplitude components and are less directly reflected in radar-observed thoracic motion. These results indicate that the proposed model preserves the main ECG temporal structure, while precise localization of subtle fiducial boundaries remains challenging.

Ablation Study. Table 2 evaluates the contribution of the main architectural components. Compared with the TCN-only baseline, introducing KI consistently

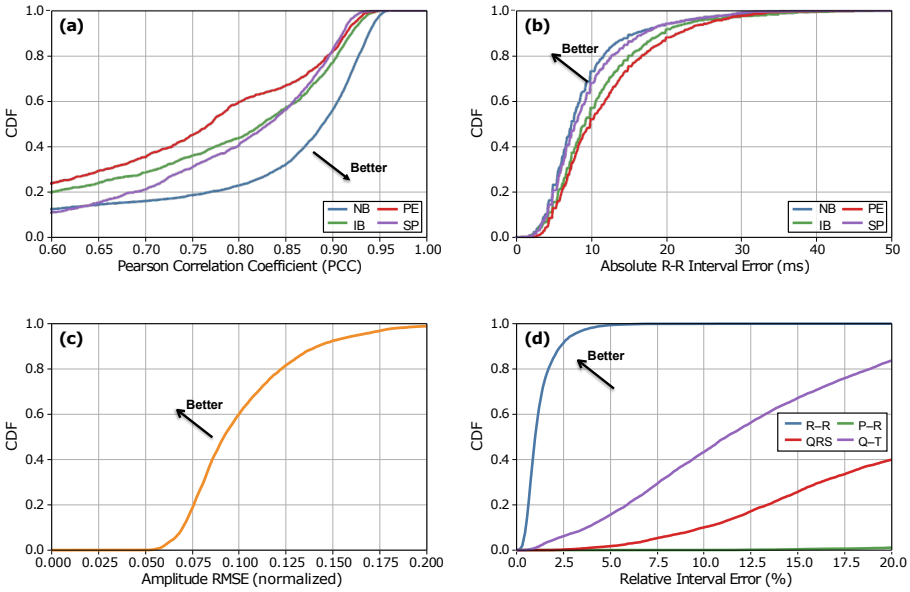


Fig. 5. CDF analysis of reconstructed ECG segments: (a) PCC, (b) R-R interval error, (c) normalized RMSE, and (d) relative clinical interval errors.

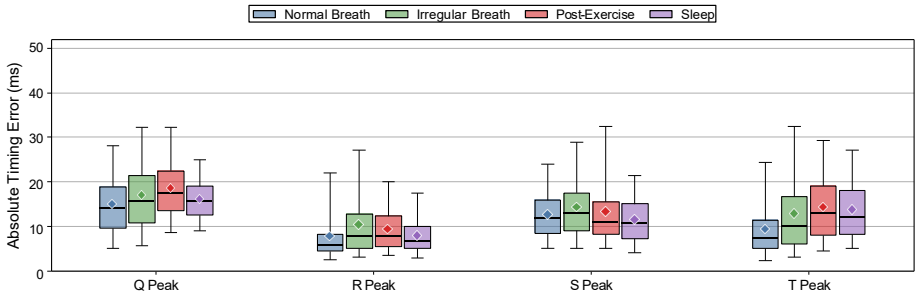


Fig. 6. Distribution of absolute fiducial timing errors for Q, R, S, and T peaks across physiological conditions. Errors are reported in milliseconds; lower values indicate better temporal alignment.

improves PCC, RMSE, and RR error, indicating that learned kinematic motion channels provide a more effective radar representation. Among the KI variants, KI16 performs better than KI8 and is therefore adopted in the final model.

CP and PA provide complementary benefits. Adding CP to the KI16-TCN baseline improves PCC from 0.730 to 0.748 and reduces RR error from 10.42 ms to 8.70 ms, showing that cardiac-prior conditioning helps guide beat-aware reconstruction. Adding PA alone mainly reduces RMSE and RR error, suggesting improved temporal alignment between radar-observed motion and ECG activity. With both CP and PA enabled, the full model achieves the best PCC, RMSE, and RR error. These results confirm that the performance gain comes from the joint use of KI, CP, and PA, rather than from the TCN backbone alone.

Table 2. Ablation study of BeatAware-Radar2ECGNet.

Variant	Params	Components			Metrics		
		KI	CP	PA	PCC↑	RMSE↓	RR_{err}↓
TCN only	1.30M	✗	✗	✗	0.683	0.111	11.79
KI8 + TCN	1.24M	8	✗	✗	0.714	0.110	10.59
KI16 + TCN	1.40M	16	✗	✗	0.730	0.109	10.42
KI16 + TCN + CP	3.18M	16	✓	✗	0.748	0.097	8.70
KI16 + TCN + PA	3.02M	16	✗	✓	0.736	0.102	10.25
Full Model	4.80M	16	✓	✓	**0.772**	**0.096**	**8.20**

Note: KI denotes the number of learned kinematic motion channels. CP and PA denote cardiac prior conditioning and physical alignment, respectively.

6 Conclusion

This paper presented the BeatAware framework for continuous ECG waveform reconstruction from FMCW radar. By combining kinematic inversion, cardiac-prior conditioning, temporal convolutional modeling, and physical alignment, the proposed framework improves waveform fidelity, fiducial timing accuracy, and beat-level monitoring reliability on MMECG. Despite these promising results, the study is limited to healthy subjects from a single public dataset, and subtle P- and T-wave recovery remains challenging. Future work will evaluate the model on broader pathological datasets and real-world sensing scenarios.

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